Chirag Deepak Shinde

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Profile

Data Engineer experienced in building end-to-end ETL pipelines and scalable data architectures for analytics. Delivered automated Azure Data Factory and Databricks pipelines that reduced manual research effort by 60% and produced reliable, model-ready datasets. Designed medallion-style Snowflake pipelines that boosted data processing efficiency by 30% and improved sales forecast accuracy by 15%. Aiming to apply data engineering expertise to drive robust, production-grade analytics solutions.

Skills

- **Data Engineering & Cloud:** Databricks, ETL/ELT, Data Modeling, Data Warehousing, Azure Data Factory, Delta Lake, Azure Synapse, Azure ADLS Gen2, Snowflake, Snowpark, Data Transformations
- **Big Data, Programming & Processing:** PySpark, Apache Spark, SQL, Python (Pandas, NumPy), Batch & Streaming Systems, Data Validation & Quality Engineering, Distributed Processing, CSV, JSON, EDI
- **Orchestration & DevOps:** Dagster, Airflow, DBT, Great Expectations, Docker, CI/CD, Git
- **Analytics & Visualization:** Power BI (DAX, Semantic Modeling), PostgreSQL, EDA
- **AI & Automation:** LLM Workflows (Gemini/OpenAI APIs), Vector Databases, RAG Architectures

Professional Experience

University of Illinois at Chicago | *Graduate Assistant* — Sep 2025 - May 2026

Project: Board Director Mortality Study & Research Outreach Automation

- Reduced manual research effort by 60% by engineering automated ETL pipelines using Azure Data Factory and Gemini Flash LLM, enabling scalable extraction of unstructured JSON and CSV web data into a structured Azure Data Lake (ADLS Gen2).
- Improved dataset reliability for research analytics by building automated data validation and data transformations with PySpark preprocessing workflows on Azure Databricks, using Unity Catalog for governance and schema control to deliver high-quality, ML-ready datasets.
- Enhanced analytical efficiency by profiling research datasets, mapping entity relationships, defining foreign key constraints, and optimizing indexing in a normalized data model - cutting query runtime and enabling direct SQL queries in Azure Synapse integrated with Power BI dashboards.
- Increased outreach workflow scalability by 50% through automating alumni and startup campaign pipelines with Power Automate, integrated with Azure SQL and other cloud data systems.

M.S Engineers | *Data Engineer* — May 2023 - Jul 2024

Project: Sales Forecasting & Analytics Data Platform

- Improved data processing efficiency by 30% by building ETL pipelines with PySpark and SQL in Snowflake using Snowpark and applying a medallion-style architecture.
- Increased sales forecast precision by 15% by creating an ingestion and transformation pipeline with Azure Event Hubs and batch processing to provide clean time-series data to Python ML models, reaching 85% accuracy.
- Enhanced analytical performance for BI teams by designing Star Schema data models and structured data marts within the organization's cloud data lake, supporting scalable reporting and KPI analysis.
- Reduced manual deployment overhead by 50% by building a containerized CI/CD DataOps pipeline using Docker, GitHub Actions and Git, automating schema validation and model retraining workflows.
- Improved production data reliability by implementing a Spark SQL quality framework with null checks, schema drift detection, statistical threshold validation, and automated A/B comparisons on live data streams - reducing pipeline failures and ensuring consistent data integrity.

Projects

RAG-based Q&A System with Vector DB — Aug 2025 - Dec 2025

- Reduced manual document search time by 70% by building an end-to-end ingestion and transformation pipeline using PySpark for distributed text processing and managing code in Git.
- Improved retrieval accuracy and filtering performance by designing structured metadata and indexing high-dimensional vectors in FAISS (HNSW/IVF) and PostgreSQL, enabling citation-grounded semantic search.
- Scaled document processing capabilities by implementing Spark-based distributed transformations to handle large-scale PDF ingestion, OCR-derived text extraction, and automated vector indexing.
- Delivered a production-ready AI application by integrating the backend RAG pipeline with Gemini LLMs for real-time querying, a responsive Streamlit frontend, and Dagster for workflow orchestration.

YouTube Mental Health Recovery Analysis — Jan 2025 - May 2025

- Automated extraction of 80+ structured features by engineering an ETL pipeline using the YouTube API, PySpark, Dagster orchestration, and RoBERTa/Gemini LLM workflows, storing processed data in a centralized research data lake.
- Improved analytical insight generation by conducting EDA and statistical modeling (Negative Binomial Regression, ANOVA, OLS) to link emotional indicators with audience engagement metrics.
- Reduced report generation time for researchers by integrating structured datasets with Power BI dashboards and writing optimized SQL queries in Python to support scalable visualization.

Education

University of Illinois at Chicago (UIC) | *Master of Science, Computer Science* — **Chicago, IL Aug 2024 - May 2026**

SPPU University | *Bachelor, Computer Engineering* — **Pune, IND Aug 2019 - May 2023**

Certifications

- Databricks Certified Data Engineer Associate | Databricks | June 2026